

Quick Activity 2: Canteen Group Order

A canteen sells rice meals at ₱85 each and drinks at ₱25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a ₱5 packaging fee per rice meal.

Decomposition

Input	Process	Output
Numbers of rice meals each, drinks each, and payment	Calculate rice meal subtotal, drink subtotal, discount on rice meals, packaging fee, discounted meal subtotal, final bill, change	Total cost and change

Pseudocode

START

 READ quantity, payment

 rice_mealsSubtotal $\leftarrow$ quantity $\times$ 85

 drinksSubtotal $\leftarrow$ quantity $\times$ 25

 discountedSubtotal $\leftarrow$ rice_mealsSubtotal $\times$ 0.10

 chargesCost $\leftarrow$ quantity $\times$ 5

 totalCost $\leftarrow$ discountedSubtotal + chargesCost $\leftarrow$ payment - totalCost

 DISPLAY totalCost, change

END

Dry run Calculation

Case 1 Input

- Rice meals = 3
- Drinks = 4
- Payment = ₱500.00

Step	Variable	Calculation	Result
1	Meal subtotal	3×85	₱255.00
2	Drink subtotal	4×25	₱100.00
3	Discount	$255 \times 10\%$	₱25.50
4	Packaging fee	3×5	₱15.00
5	Final bill	$255 - 25.50 + 100 + 15$	₱344.50
6	Change	$500 - 344.50$	₱155.50

Output

Final bill = ₱344.50

Change = ₱155.50

Test Case 2

Input

- Rice meals = 5
- Drink = 3
- Payment = ₱600.00

Step	Variable	Calculation	Result
1	Meal subtotal	5×85	₱425.00
2	Drink subtotal	3×25	₱75.00
3	Discount	$425 \times 10\%$	₱42.00
4	Packaging fee	5×5	₱25.00
5	Final bill	$425 - 42.5 + 75 + 25$	₱482.50
6	Change	$600 - 482.50$	₱117.50

Output

- Final bill = ₱482.50
- Change = ₱117.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?

- It's incorrect because it only applies to the rice meal and not in drinks. If we put the discount to drinks and packaging fee the bill decreases, causing them to lose a large amount of money.